In JEE Advanced 2016, FIITJEE Students bag 36 in Top 100 AIR, 75 in Top 200 AIR, 183 in Top 500 AIR. 3541 Students from Long Term Classroom/ Integrated School Program & 4423 Students from All Programs have qualified in JEE Advanced, 2016

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST - III

Time Allotted: 3 Hours **Maximum Marks: 360**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part has only one section: **Section-A**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 to 30)** contains 30 multiple choice questions which have only one correct answer. Each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

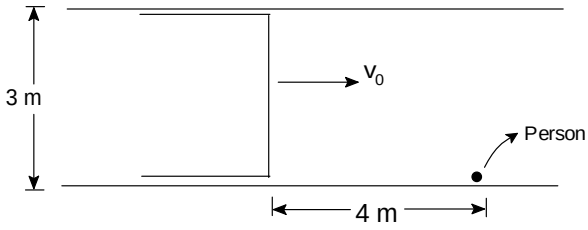
PART – I

SECTION – A

Straight Objective Type

This section contains **30 multiple choice questions** numbered 1 to 30. Each question has 4 choices (A), (B), (C) and (D), out of which only **ONE** is correct

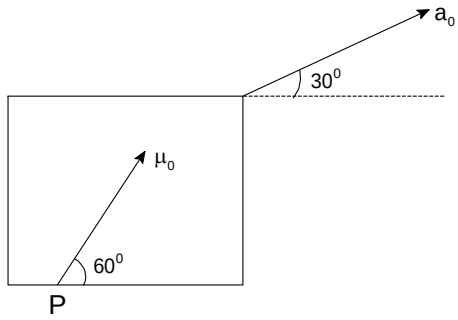
1. A '3m' wide truck is moving with a uniform velocity v_0 = 5m/s along a straight horizontal road. A person starts from rest to cross the road with an acceleration of constant magnitude 'a' m/s^2. When the truck is '4' m away from him. For what minimum value of 'a' he can cross the road safely in 2 seconds.



(A) $\frac{75}{8}\text{m/s}^2$ (B) $\frac{5}{2}\text{m/s}^2$

(C) $\frac{125}{8}\text{m/s}^2$ (D) $\frac{3}{2}\text{m/s}^2$

2. A cabin is moving in the vertical plane with constant acceleration 'a_0' as shown. A particle is projected with a velocity $u_0 = 10\,\text{m/s}$ wrt. the cabin from point 'p' as shown. For what value of 'a_0' the particle will hit point 'p' during its downward motion. $\left(g = 10\ \text{m/s}^2\right)$



(A) $\frac{10}{\sqrt{3}}\text{m/s}^2$

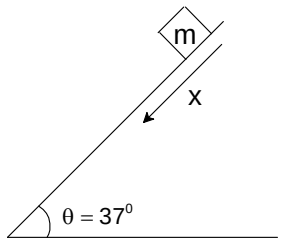
(B) $10\ \text{m/s}^2$

(C) $10\sqrt{3}\ \text{m/s}^2$

(D) All of the above

Space for rough work

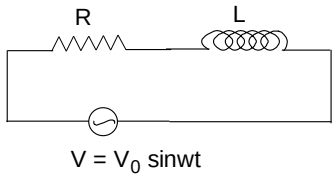
3. A block of mass 'm' starts sliding down the incline with initial zero velocity. The surface is rough and μ varies as $\mu = 0.5\,x$, where 'x' is distance travelled down the incline. What is the distance travelled by the block till the speed achieved by it is half of its maximum possible speed achieved by it during its motion?



(A) $\frac{3}{2}m$ (B) $\frac{3\sqrt{3}}{2}m$

(C) $\frac{3}{2}\left(2-\sqrt{3}\right)m$ (D) $\left(2\sqrt{3}-3\right)m$

4. In the circuit shown in figure, the energy lost in one cycle is

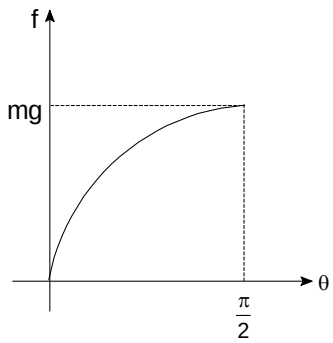


(A) zero (B) $\frac{V_0^2RT}{2R}$

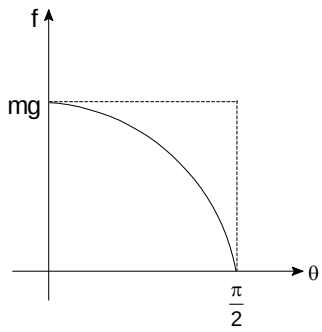
(C) $\frac{V_0^2RT}{2\left(R^2+\omega^2L^2\right)}$ (D) none of these

5. A block of mass 'm' is kept on rough inclined plane of inclination 'θ', which is varied from $\theta = 0$ to $\theta = \frac{\pi}{2}$ very slowly. Graph of friction force vs θ is given by $\left(\mu_s = \mu_k = \sqrt{2}\sin\theta\right)$

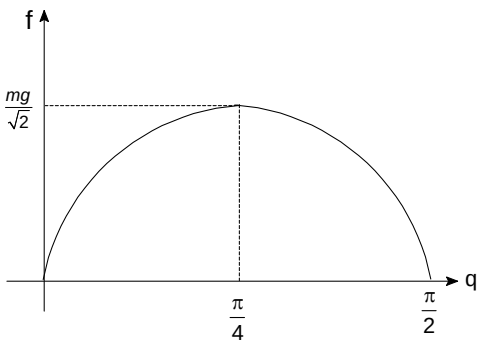
(A)



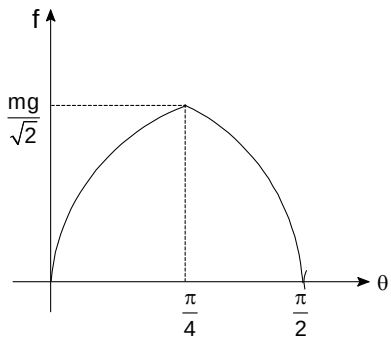
(B)



(C)



(D)



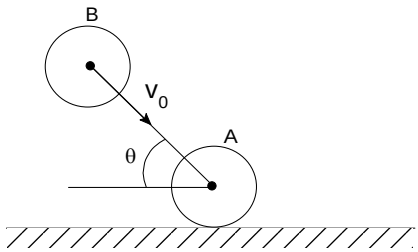
Space for rough work

6. A battery of emf E is connected in series with three resistances R, 2R and 3R. The voltage across 2R is measured with a voltmeter, whose resistance is 20 R. The percentage error (approximately) will be.

(A) –3.25% (B) –5.25%

(C) –6.25% (D) –4.25%

7. A small sphere 'A' of mass 'm' is kept on a rough horizontal surface. Another identical sphere 'B' strikes it with velocity 'v_0' at an angle 'θ' as shown. Value of 'e' for two spheres is '1' and for the sphere & surface is 'zero'. Which of the following statements is correct? $(\mu = 0.5)$



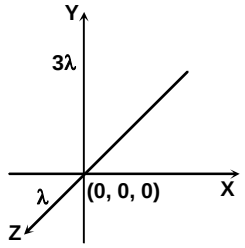
(A) Speed of sphere 'A' just after collision is $\frac{3v_0}{5}$ for $\theta = 37^0$

(B) Friction acting on the sphere during collision will be kinetic for $\theta = 30^0$

(C) Friction acting on the sphere during collision will be static for $\theta = 30^0$

(D) Nature of friction cannot be determined at $\theta = \tan^{-1}\left(\frac{4}{7}\right)$

8. The diagram shows three infinitely long uniform line charges 2λ, 3 λ and λ placed on the X, Y and Z axis respectively. The work done in moving a unit positive charge from (1, 1, 1) to (0, 1,1) is equal to

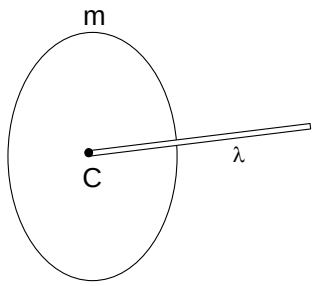


(A) $\frac{\lambda \ell n2}{2\pi \in_0}$ (B) $\frac{\lambda \ell n2}{\pi \in_0}$

(C) $\frac{3\lambda \ell n2}{2\pi \in_0}$ (D) none

Space for rough work

9. A thin ring of radius R, mass m and an infinitely long uniform wire of linear mass density λ, is placed perpendicular to the plane of the ring along the axis of the ring with one of its ends coinciding with the centre of the ring, form a system. The gravitational force of interaction between ring and the wire is



(A) $\frac{2Gm\lambda}{R}$ (B) $\frac{Gm\lambda}{R}$

(C) $\frac{Gm\lambda}{2R}$ (D) $\frac{\sqrt{2}Gm\lambda}{2}$

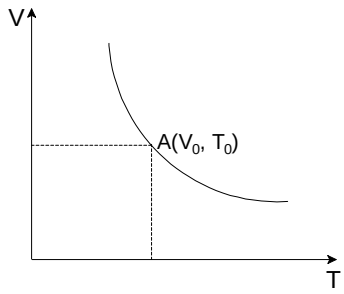
10. Consider an ideal gas with density ρ mg / cm^3 at standard pressure and temperature. If speed of sound propagation in this gas is v = 330 m/s, the degree of freedom of the gas is 5. Then the density ρ is

(A) 1.3 mg / cm^3 (B) 1.8 mg / cm^3

(C) 2.5 mg / cm^3 (D) 3 mg / cm^3

11. A gas is undergoing an adiabatic process. At a certain stage A, the value of volume and temperature is V_0 and T_0. $\left(C_v \text{ of the gas is } \frac{3R}{2}\right)$

Then $\frac{dV}{dt}$ at A, is

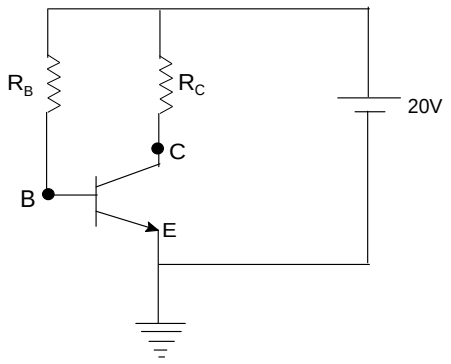


(A) $\frac{V_0}{3T_0}$ (B) $\frac{3V_0}{T_0}$

(C) $\frac{2V_0}{3T_0}$ (D) $\frac{3V_0}{2T_0}$

Space for rough work

12. For given CE biasing circuit, voltage across collector – emitter is 12 V, current gain is 100 and base current is 0.04 mA. Then the value of collector resistance 'R_c' is
(A) 1200 Ω
(B) 200 Ω
(C) 400 Ω
(D) 2000 Ω



13. Two identical metal plates with face area 'A' placed parallel to each other at separation 'd' are given charges +4Q and +6Q. If they are connected by a thin conducting wire, amount of heat produced will be given by
(A) $\frac{Q^2d}{\varepsilon_0 A}$ (B) $\frac{2Q^2d}{\varepsilon_0 A}$ (C) $\frac{5Q^2d}{2\varepsilon_0 A}$ (D) $\frac{Q^2d}{2\varepsilon_0 A}$

14. Rain drops fall from a certain height with a terminal velocity 'v' on the ground. The viscous force is $F = 6\pi\eta rv$. Here η is coefficient of viscocity, r is radius of rain drop and v is speed. Then work done by all the forces acting on the ball till it reaches the ground is proportional to
(A) r^7 (B) r^5 (C) r^3 (D) r^2

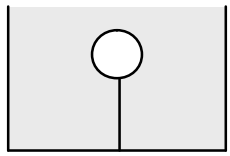
15. The threshold frequency for certain metal is f_0. When frequency of incident light is increased from $2f_0$ to $4f_0$, maximum velocity of photoelectrons is increased from v_0 to v. The value of v is
(A) $2v_0$ (B) $\sqrt{3}v_0$ (C) $4v_0$ (D) $3v_0$

16. A container has an ideal gas at pressure 'P' in thermal equilibrium. Let mass of a molecule is 'm' and all the molecules are moving with same speed 'v' in random directions. If 'n' molecules collide with unit area of container wall per second, the value of 'n' will be
(A) $\frac{P}{mv}$ (B) $\frac{P}{2mv}$ (C) $\frac{2P}{mv}$ (D) $\frac{\sqrt{3}P}{2mv}$

Space for rough work

17. A room at 20^oC is heated by a heater of resistance 20Ω connected to 200 V mains. The temperature is uniform throughout the room and the heat is transmitted through a glass window of area 1 m^2 and thickness 0.2 cm. Calculate the temperature outside thermal conductivity of the glass is 0.2 cal/ms^oC.
(A) 13.69^oC (B) 15.24^oC (C) 17.85^oC (D) 19.96^oC

18. A cork is attached to the bottom of a container filled with water with a string as shown in figure. If the container accelerate in a horizontal direction towards right. When the cork is in equilibrium w.r.t the container then which one is correct?



(A) Inclination of string with vertical is $\tan^{-1}(a/g)$ towards left.

(B) Inclination of the string with vertical is $\tan^{-1}(a/g)$ towards right.

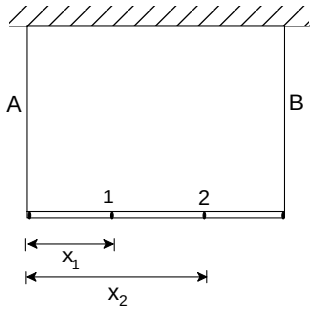
(C) Inclination of the string with vertical is $\frac{\pi}{2}-\tan^{-1}(a/g)$ towards left.

(D) Inclination of the string with vertical is $\frac{\pi}{2}-\tan^{-1}(a/g)$ towards right.

19. The temperature at the bottom of a lake of depth 'L' is 4^0C. The temperature of the air just above the layer of ice frozen at the lake's surface is -2^0C for a long time. Thermal conductivity of ice is 3 times that of water. Thickness of the frozen layer of ice should be
(A) $\frac{L}{3}$ (B) $\frac{L}{2}$ (C) $\frac{3L}{5}$ (D) $\frac{2L}{3}$

20. The de – Broglie wavelength of a photon is n times the de – Broglie wavelength of an electron. If the speed of the electron is $v_e=\frac{2c}{100n}$, then the value of $\frac{E_e}{E_p}$ is $(E \rightarrow \text{kinetic energy})$
(A) 10^{-2} (B) 1 (C) 10^2 (D) 10^{-4}

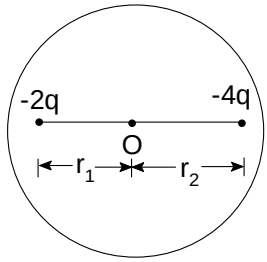
21. A light rigid rod of length 'ℓ' is suspended at its two ends by two wires 'A' and 'B' of equal lengths and area of cross section 'a' and '2a' respectively. $(Y_A=Y_0 \ \& \ Y_B=3Y_0)$
A mass 'm' is suspended at points '1' and '2', one by one. When it is suspended at point '1', stress developed in both wires is same; and when it is suspended at point '2', strain developed in both wires is same. Ratio of x_1: x_2 is given by



(A) 7 : 18 (B) 7 : 9
(C) 7 : 3 (D) 14 : 3

Space for rough work

22. Charge density for a sphere of radius R is distributed as $\rho = \rho_0 r$ for $r \le R$. Total charge on the sphere is 10q and two point charges '-2q' and '-4q' are embedded in the sphere at distances 'r_1' and 'r_2' as shown such that net force on them is zero. The value of r_2 is



(A) $\dfrac{R}{\left\{\left(\sqrt{5}\right)\left(\sqrt{2}+1\right)\right\}^{\frac{1}{2}}}$ (B) $\dfrac{R}{\left\{\left(\sqrt{10}\right)\left(\sqrt{2}+1\right)\right\}^{\frac{1}{2}}}$

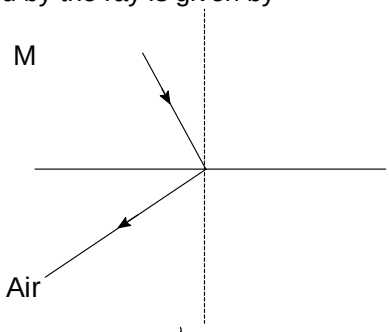
(C) $\dfrac{R}{\left\{\left(\sqrt{2}\right)\left(\sqrt{5}+1\right)\right\}^{\frac{1}{2}}}$ (D) $\dfrac{R}{\left\{\left(\sqrt{2}\right)\left(\sqrt{2}+1\right)\right\}^{\frac{1}{2}}}$

23. For a glass prism $\left(\mu = \sqrt{2}\right)$, the angle of minimum deviation is equal to the angle of the prism. The angle of the prism is equal to

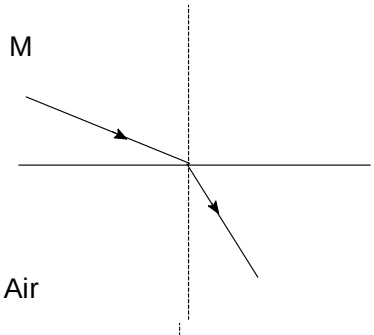
(A) 30^0 (B) 90^0 (C) 60^0 (D) 75^0

24. A certain material has negative refractive index. A ray is incident from this medium into air. Path followed by the ray is given by

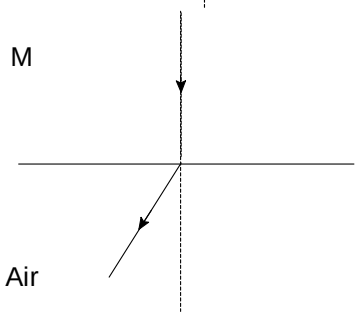
(A)



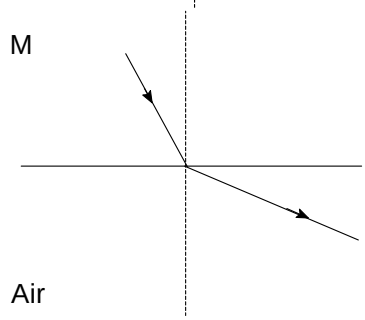
(B)



(C)

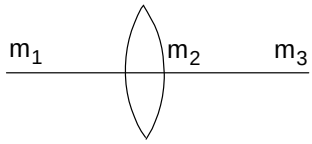


(D)



Space for rough work

25. The figure shows an equi-convex lens. What should be the condition of the refractive indices so that the lens become diverging



(A) $2\mu_3 > \mu_1 - \mu_2$ (B) $2\mu_2 < \mu_1 + \mu_3$

(C) $2\mu_2 > 2\mu_1 - \mu_3$ (D) none of these

26. A resistance coil of resistance 'r' connected to an external battery is placed inside an adiabatic cylinder fitted with a frictionless piston of same area. Cylinder contains 2 mol of ideal gas O_2. Current flows through the coil such that temperature of the gas varies as $T = T_0 + \alpha t + \beta t^2 + \gamma t^3$, keeping pressure constant with time 't'. Current flowing through the coil at t = 1s is given by

$\left(\text{where } \alpha = 1\,s^{-1}, \beta = \frac{1}{2}s^{-2}, \gamma = \frac{1}{3}s^{-3}\right)$

(A) $\sqrt{\frac{7R}{r}\left(T_0 + \frac{7}{6}\right)}$ (B) $\sqrt{\frac{21R}{r}}$

(C) $\sqrt{\frac{7R}{r}\left(\frac{7}{6}\right)}$ (D) $7\sqrt{\frac{R}{r}}$

27. A certain radionuclide is known to become weaker in activity by 5% every day. Mean life of the radionuclide will be (Approximately)

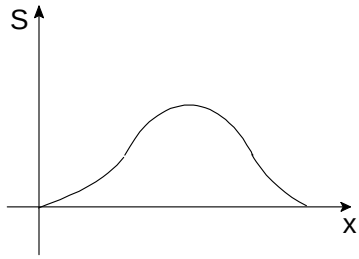
(A) 20 days (B) 20 hours (C) 20 min (D) 20 sec

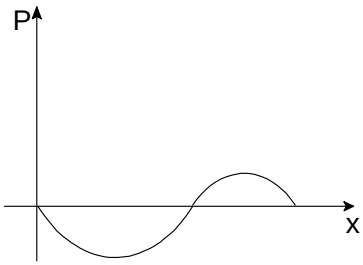
28. A $\beta-$ particle is moving along a circle of radius 'R' with a constant angular velocity 'ω'. Point 'A' lies in the same plane at a distance '$\frac{2}{\sqrt{3}}R$' from the centre. Point 'A' records magnetic field produced by the $\beta-$particle. If the minimum time interval between two successive times at which 'A' records zero magnetic field is 't', 'ω' in terms of t' is given by

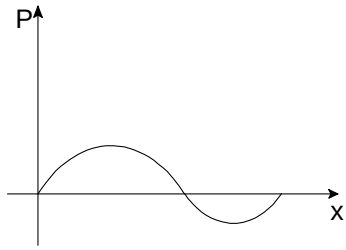
(A) $\frac{2\pi}{3t}$ (B) $\frac{\pi}{3t}$ (C) $\frac{3\pi}{4t}$ (D) $\frac{\sqrt{3}\pi}{2t}$

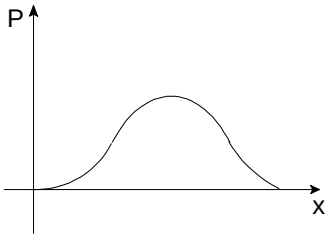
Space for rough work

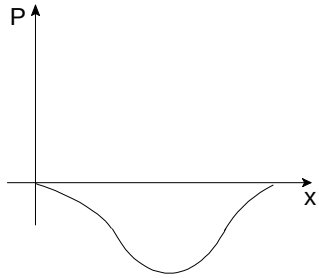
29. For the given displacement (s) vs 'x' graph for a sound wave, appropriate excess pressure (P) vs 'x' graph is



(A)


(B)


(C)


(D)


30. If between wavelength λ and $\lambda + d\lambda$ e_λ and a_λ are the emissive power and absorptive powers of a body and E_λ is the emissive power of a perfectly black body, then according to Kirchhoff's law which is true?

(A) $e_\lambda = a_\lambda = E_\lambda$ (B) $e_\lambda E_\lambda = a_\lambda$

(C) $e_\lambda = a_\lambda E_\lambda$ (D) $e_\lambda a_\lambda E_\lambda = constant$

Space for rough work

Chemistry

PART – II

SECTION – A

Straight Objective Type

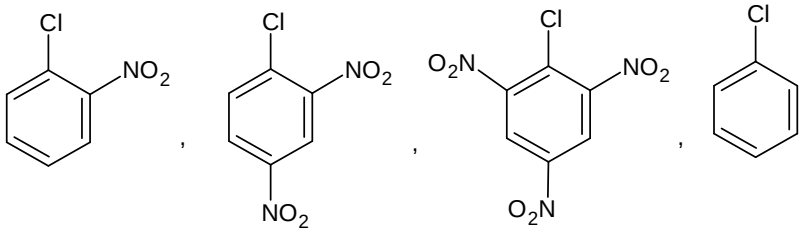
This section contains **30 multiple choice questions** numbered 1 to 30. Each question has 4 choices (A), (B), (C) and (D), out of which only **ONE** is correct

1. Which of the following pair from the same osazone derivative?
(A) D – Glucose and L - glucose (B) D – Glucose and D - fructose
(C) D – Glucose and D galactose (D) None of these

2. $CH_3 \equiv CCH_3 \xrightarrow[\text{Lindlar's Catalyst}]{H_2} A \xrightarrow{Br_2} B$. Then B is
(A) cis (B) trans (C) meso (D) racemic

3. Nineteen grams of molten $SnCl_2$ is electrolyzed for some time using inert electrodes until 0.119 g of Sn is deposited at the cathode. No substance is lost during the electrolysis. Find the ratio of the masses of $SnCl_2$ and $SnCl_4$ after electrolysis.
(A) 65.71 (B) 71.62 (C) 85.4 (D) None

4. Arrange the following compounds in increasing order of reactivity towards nucleophilic substitution & choose correct option



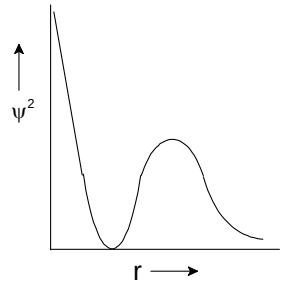
(A) (B) (C) (D)

(A) A > B > C > D (B) C > B > A > D (C) B > A > C > D (D) None

5. Buna – s rubber is a polymer of
(A) $H_2C{=}C(CH_3){-}CH{=}CH_2$ (B) $H_2C{=}C(Cl){-}CH{=}CH_2$
(C) $CH_2 = CH - CH = CH_2$ & $CH_2 = CHCN$ (D) $CH_2 = CH - CH = CH_2$ & $C_6H_5CH = CH_2$

Space for rough work

6. The value of $(n+\ell)$ (where n = principle quantum no and ℓ = azimuthal quantum no) for the orbital shown by following curve



(A) 3 (B) 4
(C) 2 (D) None

7. A radioactive isotope ${}_{Z}A^{m}$ ($t_{1/2}$ = 10 days) decays to give ${}_{Z-6}B^{m-12}$ stable atom along with α – particle. If m gram of 'A' are taken & kept in a sealed tube. How much He will accumulate in 20 days. (calculate amt. in lit assuming process is occurring at STP).
(A) 22.4 lit (B) 2.25 lit (C) 50.4 lit (D) 100.8 lit

8. In 1 L saturated solution of AgCl $\left[K_{sp}(AgCl) = 1.6\times10^{-10}\right]$, 0.1 mole of CuCl $\left[K_{sp}(CuCl) = 1.0\times10^{-6}\right]$ is added. The resultant conc. of Ag^{+} in the solution is 1.6×10^{-x} and y = 10 – x, then the value of y is
(A) 4 (B) 7 (C) 3 (D) None

9. In seven possible crystal system how many crystal systems have $\alpha = \beta$?
(A) 2 (B) 4 (C) 5 (D) 7

10. The distance between asoctahedral & tetrahedral void in fcc lattic would be
(A) $\sqrt{3}a$ (B) $\frac{\sqrt{3}a}{2}$ (C) $\frac{\sqrt{3}a}{3}$ (D) $\frac{\sqrt{3}a}{4}$

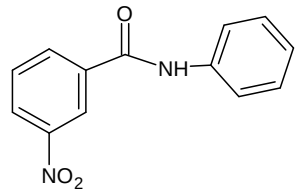
11. 0.1 M KI and 0.2 M AgNO3 are mixed in 3 :1 volume ratio. The depression of freezing point of the resulting solution will be $\left[K_f(H_2O) = 1.86\ K\ Kg\ mol^{-1}\right]$.
(A) 3.72 K (B) 1.86 K (C) 0.93 K (D) 0.279 K

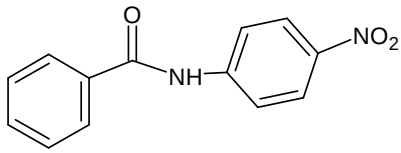
12. An e^{-} moves from the $n = 5$ to $n = 1$ energy level in H like atom and emits photons of energy 96J. Then the ionization energy of the given species is:
(A) -96 J (B) +96 J (C) 100 J (D) None

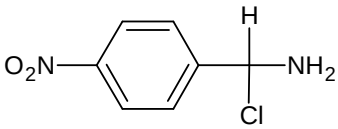
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13. $(P) \xrightarrow[(ii)\ H^+]{(i)\ NaOI} (Q) \xrightarrow{SOCl_2} (R) \xrightarrow[NaOH]{C_6H_5NH_2} (S) \xrightarrow[Conc.H_2SO_4\ (1eq)]{Conc.HNO_3} T(major)$

T can be:

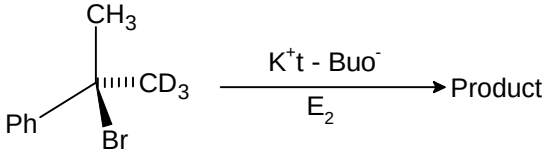
(A) 

(B) 

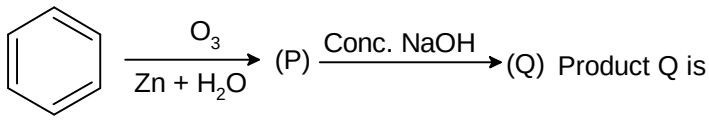
(C) 

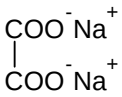
(D) None

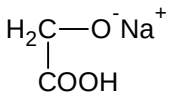
14. Which best described the product of the following reaction?

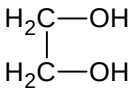


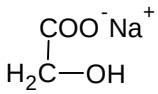
(A) Absolute configuration has been inverted (B) Absolute configuration has been retained
(C) Racemization has occured (D) Loss of chirality has occured

15.  Product Q is

(A) 

(B) 

(C) 

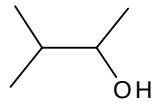
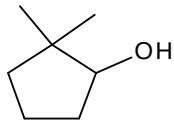
(D) 

Space for rough work

16. What happens when chromium hydroxide is treated with hydrogen peroxide in the presence of NaOH?
(A) yellow coloured sodium chromate is obtained
(B) Blue coloured sodium chromate is obtained
(C) Cr^{+3} reduces to Cr^{+2}
(D) orange solution is obtained

17. Which of the following is microcosmic salt?
(A) $NaHCO_3$ (B) $Na_2S_2O_3$ (C) $Na(NH_4)HPO_4$ (D) $Na_2CO_3.10H_2O$

18. Which of the following equations is involved in Solvay process?
(A) $NaCl + NH_3 + H_2O + CO_2 \longrightarrow NH_4Cl + NaHCO_3$
(B) $CaO + 2NH_4Cl \longrightarrow 2NH_3 + H_2O + CaCl_2$
(C) $2NaHCO_3 \longrightarrow Na_2CO_3 + CO_2 + H_2O$
(D) All

19. (1)  $\xrightarrow[\Delta]{H^+}$ (P) (Major) (2)  $\xrightarrow[\Delta]{H^+}$ (Q) (Major)
The total number of α - hydrogens in P + Q is
(A) 13 (B) 15 (C) 17 (D) 19

20. When NO and NO_2 are mixed the following equilibrium are readily obtained
$2NO_2 \rightleftharpoons N_2O_4$ $K_p = 6.8\ atm^{-1}$
and $NO + NO_2 \rightleftharpoons N_2O_3$
In an experiment when NO and NO_2 are mixed in the ratio of 1 : 2, the final total pressure was 5.05 atm and the partial pressure of N_2O_4 was 1.7 atm. Calculate the equilibrium partial pressure of NO.
(A) 2.5 atm (B) 1.8 atm (C) 1.05 atm (D) None

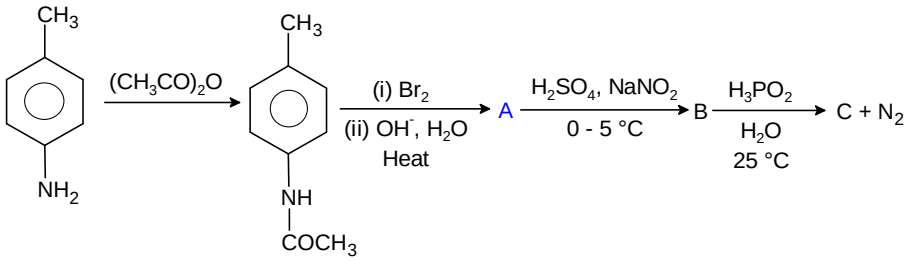
Space for rough work

21. Which of the following can be most readily, liquefied
$(\text{Give} - \text{value of 'a' for } NH_3 = 4.17, CO_2 = 3.59, SO_2 = 6.71, Cl_2 = 6.49)$
(A) NH_3 (B) Cl_2 (C) SO_2 (D) CO_2

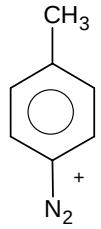
22. 1 mole of an ideal gas $A\left(C_{v.m} = 3R\right)$ and mole of an ideal gas B are $\left(C_{v.m} = \frac{3}{2}R\right)$ taken in a container and expanded reversibly & adiabatically from 1 lit to 4 lit starting from initial temperature of 320 k. ΔE or ΔU for process is
(A) -240 R (B) -205 R (C) -405 R (D) None

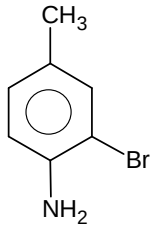
23. Magnetic behavior of complex $\left[Fe(en)_2 Cl_2\right]Cl$ (en = ethane -1, 2-diamine) is
(A) Paramagnetic with 2 unpaired e^- (B) diamagnetic
(C) paramagnetic with one unpaired e^- (D) None

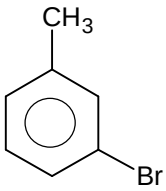
24. For reaction



Then C is

(A) 

(B) 

(C) 

(D) None

Space for rough work

25. In sp^3d^3 hybridisation, which orbitals are involves?
(A) $dx^2 - y^2$, dz^2, dxy (B) dxy, dyz, d_2x
(C) $dx^2 - y^2$, dxy, dxz (D) dz^2, dyz, dzx

26. Which of following reaction does not occur in Bessemer converter?
(A) $2Cu_2S + 5O_2 \longrightarrow 2CuSO_4 + 2CuO$
(B) $2Cu_2S + 3O_2 \longrightarrow 2Cu_2O + 2SO_2 \uparrow$
(C) $2CuFeS_2 + O_2 \longrightarrow Cu_2S + 2FeS + SO_2$
(D) $FeO + SiO \longrightarrow FeSiO_3$

27. Consider the following reaction
$2XS + 3O_2 \xrightarrow{\Delta} 2XO + 2SO_2$
$2XO + XS \xrightarrow{\Delta} 3'X' + SO_2$
'X' can not be
(A) Hg (B) Pb (C) Zn (D) None

28. In the series reaction -
$A \xrightarrow{K_1} B \xrightarrow{K_2} C \xrightarrow{K_3} D.$
If $K_1 > K_2 > K_3$ then the rate determine step of the reaction is
(A) $A \rightarrow B$ (B) $C \rightarrow D$
(C) $B \rightarrow C$ (D) any step

29. $FeCr_2O_4 + NaOH + air \longrightarrow (A) + Fe_2O_3$
$(A) + (B) \longrightarrow Na_2Cr_2O_7$
Then compound (A) and (B) are
(A) Na_2CrO_4, H_2SO_4 (B) $Na_2Cr_2O_7$, HCl
(C) Na_2CrO_3, H_2SO_4 (D) None

30. For reaction:
$Na_2Cr_2O_7 + X \xrightarrow{\Delta} Cr_2O_3$
$Cr_2O_3 + Y \xrightarrow{\Delta} Cr$
(X) and (Y) are
(A) C and Al (B) Al and C (C) C in both (D) Al in both

Space for rough work

Mathematics PART – III

SECTION – A

Straight Objective Type

This section contains **30 multiple choice questions** numbered 1 to 30. Each question has 4 choices (A), (B), (C) and (D), out of which only **ONE** is correct

1. The number of ways in which 10 persons can sit in a row having 40 chairs such that at least two chairs remain vacant between any two persons is
(A) $^{22}C_{10}10!$ (B) $^{22}C_{10}12!$ (C) $^{40}C_{10}10!$ (D) $^{49}C_{10}10!$

2. For any G.P, S_K denotes sum of first 'K' terms, the quantities $S_{2n}, 2(S_{2n}-S_n)$ and $S_{3n}-S_n$ always from
(A) G.P (B) A.P (C) H.P (D) none of these

3. If $\frac{\alpha^2+3\alpha-4}{2}=\frac{2\beta^2-3\beta-2}{1}=K$, then the value of $\left|\frac{2-\alpha\beta}{2K}\right|$ is equal to
(A) $\frac{1}{2}$ (B) 2 (C) 4 (D) $\frac{1}{4}$

4. The number of solution for 'θ' in the equation $\sum\left[(-1)^{q+r}\frac{n!}{p!q!r!}\sin^{2q}\theta.\cos^{2r}\theta\right]=1$, where p + q + r = n, p, q, r $\in$ N is
(A) 0 (B) 2
(C) 4 (D) can't be solved manually

5. The inverse of the contrapositive of the conditional statement $p \rightarrow \sim q$ is equivalent to
(A) the contrapositive of the inverse of $p \rightarrow \sim q$ (B) negation of $p \rightarrow \sim q$
(C) negation of the converse of $p \rightarrow \sim q$ (D) none of these

6. If the domain of the function g(x) is (0, 1) then the domain of the function $y=g\left(\frac{1}{2^x-1}\right)+g(\ln|x|)$
(A) $(0,\infty)$ (B) $(0,1)$ (C) $(1,\infty)$ (D) None of these

Space for rough work

7. If the inequality $\frac{mx^2+3x+4}{x^2+2x+2}<5$ is satisfied for all $x \in R$, then

(A) $1 < m < 5$ (B) $-1 < m < 1$ (C) $-5 < m < \frac{11}{24}$ (D) $m < \frac{71}{24}$

8. The value of $\sum_{n=1}^{\infty}\left[\frac{1+2^n}{n!}(\ln 2)^n\right]$ is

(A) $e^2 - 2$ (B) e (C) 4 (D) 2^e

9. If the lines $x = a + m$, $y = -2$ and $y = mx$ are concurrent, then the least value of $|a|$ is

(A) 0 (B) $\sqrt{2}$ (C) $2\sqrt{2}$ (D) 2

10. Let f(x) be a continuous function on [0, 4] satisfying $f(x)f(4-x) = n^{-2}$. The value of the definite integral $\int_0^4 \frac{1}{1+nf(x)}dx$ is

(A) 0 (B) 1 (C) 2 (D) 4

11. Let AB be a line segment of length 'd' units with A on the line $y = x$ and B on the line $y = -x$. The locus of the midpoint of all such line segments

(A) circle (B) parabola (C) hyperbola (D) ellipse

12. If $a_1 = 1$ and $a_n = n(1+a_{n-1}) \forall\, n \geq 2$, then the value of $\lim_{n\to\infty}\left(1+\frac{1}{a_1}\right)\left(1+\frac{1}{a_2}\right)\left(1+\frac{1}{a_3}\right)....\left(1+\frac{1}{a_n}\right) =$

(A) e (B) ln 2 (C) $e^{\frac{1}{2}}$ (D) $\frac{1}{\ln 2}$

13. The function f defined by $f(x) = \lim_{x\to\infty}\left\{\frac{(1+\sin\pi x)^t - 1}{(1+\sin\pi x)^t + 1}\right\}$ is

(A) continuous in $(-\infty, \infty)$ (B) discontinuous at all integral values of x

(C) continuous at x = 0 only (D) none of these

Space for rough work

14. If α, β, γ are the roots $px^3 + qx^2 + r = 0$, the value of the determinant $\begin{vmatrix} \alpha\beta & \beta\gamma & \gamma\alpha \\ \beta\gamma & \gamma\alpha & \alpha\beta \\ \gamma\alpha & \alpha\beta & \beta\gamma \end{vmatrix}$ is

(A) pq (B) qr (C) 0 (D) pr

15. If $A = \begin{bmatrix} 0 & 1 & 2 \\ 0 & 0 & 3 \\ 0 & 3 & 1 \end{bmatrix}$, then which of the following is true?

(A) $A^2 - A + 9I = 0$ (B) $A^2 - A - 9I = 0$
(C) $A^3 - A^2 - 9A = 0$ (D) $A^3 - A^2 + 9A = 0$

16. Vectors $\vec{a}, \vec{b}, \vec{c}$ are such that every pair is non-collinear and the vector $\vec{a} + \vec{b}$ is collinear with $\vec{c}$ and the vector $\vec{b} + \vec{c}$ is collinear with $\vec{a}$. The resultant of $\vec{a} + 2\vec{b} + 3\vec{c}$ is

(A) null vector (B) $2\vec{b}$ (C) $3\vec{c}$ (D) none of these

17. A plane meets the coordinate axes in A, B and C points such that the centroid of ΔABC is the point (p, q, r). The equation is the plane is

(A) $\frac{x}{p} + \frac{y}{q} + \frac{z}{r} = 1$ (B) $\frac{x}{2p} + \frac{y}{2q} + \frac{z}{2r} = 1$

(C) $\frac{x}{3p} + \frac{y}{3q} + \frac{z}{3r} = 1$ (D) $\frac{3x}{p} + \frac{3y}{q} + \frac{3z}{r} = 1$

18. A and B are two independent witnesses (i.e. there is no collusion between them) in a case. The probability that A will speak the truth is 'x' and the probability that B will speak the truth is 'y'. A and B agree in a certain statement. The probability that the statement is true is

(A) $\frac{x - y}{x + y}$ (B) $\frac{xy}{1 + x + y + xy}$

(C) $\frac{x - y}{1 - x + y + 2xy}$ (D) $\frac{xy}{1 - x - y + 2xy}$

Space for rough work

19. A random variable x takes the value of 0, 1, 2, 3, … with probability proportional to $(x+1)\left(\frac{1}{5}\right)^x$. Then

(A) $p(x=0)=\frac{9}{25}$ (B) $p(x\geq 1)=\frac{9}{25}$

(C) $p(x\leq 1)=\frac{16}{25}$ (D) none

20. $\int \frac{1}{\sqrt{1+x^2}\left(\sqrt{x^2+1}-x\right)^{15}}dx = A\left\{x+\sqrt{1+x^2}\right\}^n + C$, then

(A) $A=\frac{1}{14}, n=14$ (B) $A=\frac{1}{15}, n=15$

(C) $A=-\frac{1}{16}, n=-16$ (D) $A=-\frac{1}{15}, n=-15$

21. If $I=\int_0^1 \frac{dx}{1+x^{\frac{\pi}{2}}}$, then

(A) I > ln 2 (B) I < ln 2 (C) $I>\frac{\pi}{4}$ (D) none of these

22. Let $f(x)=\begin{cases}1-|x|, & |x|\leq 1\\ 0, & |x|>1\end{cases}$, $g(x)=f(x-1)+f(x+1)\ \forall\ x\in R$ then g(x) is

(A) odd only (B) even only
(C) neither odd or even (D) None of these

23. Area bounded by the curves $y=\left[\frac{x^2}{64}+2\right]$, y = x – 1, x = 0 and $y=\sin^{-1}\left[\frac{|x|}{4}\right]$ is ([.] denotes the greatest integer function)

(A) 2 (B) 3 (C) 4 (D) $2\sqrt{3}$

Space for rough work

24. Solution of the differential equation $2y\sin x.\frac{dy}{dx} = \sin 2x - y^2\cos x$ satisfying $y\left(\frac{\pi}{2}\right) = 1$

(A) $y^2 = \sin x$ (B) $y = |\sin x|$

(C) $y^2 = \cos x + 1$ (D) $y^2\sin x = 4\cos^2 x + 1$

25. If $f'(1) = 2$ and $g'\left(\sqrt{2}\right) = 4$, then the derivative of $f(\tan x)$ with respect to $g(\sec x)$ at $x = \frac{\pi}{4}$ is

(A) 1 (B) $\sqrt{2}$ (C) $\frac{1}{\sqrt{2}}$ (D) 2

26. If the imaginary part of the expression $\frac{z-1}{e^{i\theta}} + \frac{e^{i\theta}}{z-1}$ be zero, then the locus of 'z' can be

(A) a straight line parallel to the x-axis (B) a parabola
(C) a circle of unity radius (D) None of these

27. If $\alpha = \frac{2\pi}{7}$, then the value of $\tan\alpha.\tan 2\alpha + \tan 2\alpha.\tan 4\alpha + \tan 4\alpha.\tan\alpha$ is

(A) – 7 (B) – 4 (C) 0 (D) 4

28. If $\frac{x^2}{3} - 4x + 13 = \sin\left(\frac{a}{x}\right)$ for some real value of 'x' then 'a' is equal to

(A) $(2n+1)\frac{\pi}{2}$ (B) $3(4n+1)\frac{\pi}{2}$ (C) $3(1+4n)\pi$ (D) None of these

29. If the parabola y = f(x), having axis parallel to y-axis, touches the line y = x at (1, 1) then

(A) $2f'(0) + f(0) = 1$ (B) $2f(0) + f'(0) = 1$ (C) $2f(0) - f'(0) = 1$ (D) $2f'(0) + f(0) = 1$

30. If $x_1, x_2, x_3 \ldots, x_{10}$ are observations such that the sum of the algebraic deviations measured about '4' is 10 and the sum of their squares is 100. The triangle formed by the sides equal to 4, standard deviation and the mean of the sample is

(A) equilateral (B) scalene triangle (C) isosceles triangle (D) None of these

Space for rough work